

Chapter

Telluric Electric Frequency Selection Method (TEFSM) in Geophysical Groundwater Exploration: Emerging Issues

Modreck Gomo and Thandeka Ngobe

Abstract

The Telluric Electric Frequency Selection Method (TEFSM) is a surface geophysical approach that measures the horizontal component of the natural electric field. Most of the TEFSM equipment records the electric potential difference (EPD) at 5 m depth internal. Due to its low purchase and operation cost, the TEFSM has been favourably received in many developing regions, such as Africa. While the TEFSM approach is already offering an alternative but also complementary solution to the conventional geophysical methods for groundwater exploration, some emerging issues are noteworthy. This chapter highlights emerging issues of interest and discusses the challenges and positive experiences from the application of the TEFSM approach in groundwater exploration. Studies have reported various successes in groundwater prospecting and other hydrogeological investigations. Depth capabilities, portability, real-time, low purchase, and low operation cost make the TEFSM a promising approach. However, the lack of access to scientific guidance about the principles of the approach and data analysis and interpretation procedures is a challenge. Knowledge of the effect of noise on the TEFSM measurements is needed for quality control purposes. More research is needed to develop the standard approaches for analysing and interpreting the TEFSM data.

Keywords: aquifers, borehole siting, geophysics groundwater exploration, groundwater detector, hydrogeology, telluric electric frequency selection method (TEFSM)

1. Introduction

Groundwater prospecting involves the location of sites with good groundwater potential for drilling and abstraction/production boreholes. Good groundwater potential implies sites with the best chance to intersect aquifers. Geophysical surveys are the most conventional approaches used to locate sites with good groundwater potential. Electromagnetics and electrical resistivity are some of the common traditional geophysical methods that have been used successfully for

ground exploration [1–8]. However, due to the high cost, access to equipment is limited in many developing countries. The recently developed Telluric Frequency Selection Method (TEFSM) offers an alternative low-cost [9, 10].

The TEFSM approach measures the horizontal component of the electric field resulting from the Earth's natural telluric field [11, 12]. The use of a natural electric field strength observed at different frequencies for groundwater prospecting was developed by Chinese scientists in the late 70s [9]. The Earth's electrical properties at depths are the result of different frequencies of the electric field; hence, the approach is referred to as the TEFSM.

Due to its low cost, compatibility, and rapidness of field measurements among other factors, the approach has been favourably received, particularly in developing countries [13, 14]. The technology has already been applied successfully in some hydrogeology settings [10, 11, 14, 15]. However, despite studies reporting the success of the TEFSM approach in groundwater studies, several knowledge gaps exist. The lack of a standard approach to analysing and interpreting TEFSM data is a challenge. Due to this challenge, it is therefore difficult to consistently apply the TEFSM approach or evaluate the results against the standard protocol. There also appear to be different schools of thought about the principles of the TEFSM method. Some studies suggest that the TEFSM evaluates the electrical resistivity of the subsurface geological materials [10, 16], while others suggest the shielding effect phenomenon [9]. Somehow, it does appear that the application of TEFSM technology might have gone ahead of the scientific required knowledge to guide its meaningful application. A discussion of emerging issues on the application of the TEFSM in groundwater exploration is therefore imperative.

The study starts by exploring factors that could have led to a situation where the introduction and the application of the TEFSM went ahead of the scientific guidance. This analysis is partially based on the author's experience and literature review. The focus of the chapter is then shifted to present and discuss the positives and challenges encountered during the application TEFSM approach in groundwater exploration. Implications for field applications and potential future research are then highlighted. This information is important at this stage to provide a broader overview of the current understanding of the TEFSM field application to groundwater practitioners. It is expected that this discussion will stimulate research to improve the knowledge on the principles, data processing, presentation, analysis, and interpretation to optimise the application of the TEFSM approach in groundwater studies.

2. Theoretical background

The literature appears to present two schools of thought on the underlying principles of the TEFSM. Some studies suggest that subsurface resistivity or electrical conductivity of the subsurface is the main property influencing the TEFSM electric potential difference (EPD) measurements [10, 12, 13, 15, 17, 18].

In contrast, one of the technology's manufacturers, Ref. [9] reports that the TEFSM measurements reflect the shielding effectiveness of the subsurface geology to the telluric electric field waves. The shielding effectiveness indicates the subsurface geology's ability to absorb the induced secondary electric field waves. When compared to rocks, water has a higher shielding effectiveness due to its polar molecule composition. Hence, saturated rocks absorb more electric field waves, resulting in low EPD

readings being recorded, which is inferred to be indicative of groundwater availability potential. The groundwater availability potential relates to the likelihood of intercepting saturated rocks and not the quality of the groundwater. In principle, saturated rocks can be aquifers or aquitards; hence, there is an uncertainty associated with the approach. Often, weathered rocks and geological discontinuities can store and transmit the water; hence, they are potential groundwater targets for groundwater exploration. Hunan Puqi [9] is one of the technology manufacturers that has provided a detailed report with the principles and hardware components of the TEFSM approach. The underlying principles provided in Ref. [9] are used in this discussion. More research is, however, recommended to provide more clarity between the TEFSM and resistivity approach in both theory and case study field applications.

The TEFSM approach measures the horizontal component of the electric field generated by the telluric Electromagnetic (EM) field in the audio frequency range. The telluric EM field has three main natural sources, namely [12]:

- Outside the earth, which is like magnetotellurics [19–21]
- Earth's inner mantle.
- Stray industrial currents.

When considerations are made at a site far away from the source, the distribution of the telluric EM field from the three sources is in horizontal components (magnetic and electrical). Hence, the horizontal components of magnetic (E_x) and electrical (E_x) can be considered as incidental plane waves perpendicular to the ground surface, and their propagation is governed by Maxwell's equations (Eqs. (1) and (2)).

$$\nabla^2 H + k^2 H = 0 \quad (1)$$

$$\nabla^2 E + k^2 E = 0 \quad (2)$$

E (V/m) and H (A/m)—electric and magnetic fields, k —wave number, where:

$$k = [\omega^2 \mu \epsilon - i \omega \sigma \mu]^{1/2} \quad (3)$$

μ (H/m)—the magnetic permeability, ω (rad s⁻¹)—angular frequency, σ (S/m)—electric conductivity (where electric resistivity $\rho = 1/\sigma$ (Ω m)), ϵ (F/m)—dielectric permittivity.

When the earth is considered uniform, the surface impedance (Z) (Eq. (4)) can be related to the frequency of the telluric EM waves, E_x , H_x , and the earth's apparent resistivity (Eq. (5)).

Where:

$$Z = \sqrt{\omega \rho \mu \epsilon^{i\pi/4}} \quad (4)$$

$$\rho = \frac{1}{5f} |Z|^2 = \frac{1}{5f} \left| \frac{E_x}{H_x} \right|^2 \quad (5)$$

When a local-scale area of investigation is considered, the H_x is regarded to be stable and hence constant [9]. The qualitative relationship between E_x and ρ can thus be used to investigate the geological rock's electrical properties [9].

The skin depth (δ) of the investigation is computed using the f and ρ (Eq. (6)).

$$\delta \approx 503 \sqrt{\frac{\rho}{f}} \quad (6)$$

In the TEFSM, the resistivity of the underground geology at the point of measurement is regarded as constant, and Eq. (6) shows that a large investigation depth is realised at low frequency. When the resistivity is taken as a constant, the operating frequency is varied to change the investigation depth [9].

Furthermore, research has revealed that [9]:

- “Observation on the surface of the earth for the strength of telluric electric field brings information at certain depths of the earth’s electric distribution.” The depth of investigation is determined by the electric field wave frequency when resistivity is assumed constant (Eq. (6)).
- “When the magnetic field is assumed to be constant at local-scale investigation site, multiple-frequency electric field values are only observed on the earth’s surface.”
- “The varying frequencies of the electric field reflect the earth’s electrical properties at different depths.”
- When measurements are made for the electric field only, computing the earth structure’s resistivity is not possible. It is therefore only possible to make a qualitative evaluation of resistivity by assessing the shielding effect of the earth’s underground geological rocks on the electrical field.

It is important to emphasise that the TEFSM approach does not evaluate the resistivity of the earth’s underlying materials but its shielding effect on telluric electric waves, which is largely controlled by the presence of water molecules. According to Ref. [22], *“It is clear that the groundwater detector values, and resistivity are two completely different concepts, and groundwater detector values cannot be converted to resistivity values.”*

2.1 TEFSM equipment and data collection

A schematic basic field setup of the TEFSM equipment is shown in **Figure 1**. Electric field measurements at different frequencies corresponding to different depths are recorded as the electrical potential difference (V) between two grounded copper electrodes [9]. A 10 m electrode spacing is recommended for optimum spacing [23]. The station of measurement is located at the centre of the two electrodes. The basic TEFSM equipment only makes one measurement at a station and time. Electrodes are moved along the survey line to make the measurements at subsequent stations. The advanced TEFSM technology has multiple electrodes and a much deeper rated investigation depth of up to 3000 m, for example [24]. This makes it possible to obtain multiple measurements without moving the electrodes.

The TEFSM meter records the EPD at a depth interval. The depth internals of measurement will likely vary among models or manufacturers. However, the 5 m depth interval appears to be more common. The recorded data is displayed on a monitor that offers a platform for real-time data viewing (**Figure 1**) in the field.

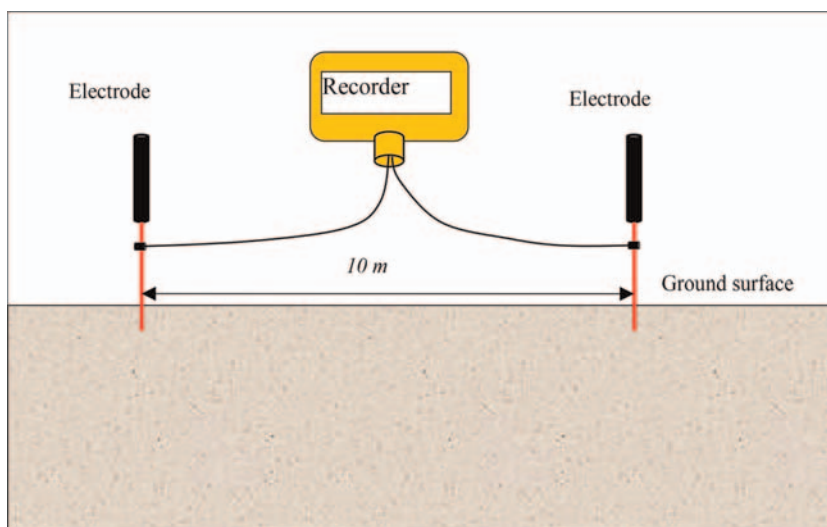


Figure 1.
Schematic basic field setup of TEFSM equipment.

3. TEFSM data processing and analysis

There are different approaches to processing and analysing TEFSM data. The first approach uses in-built software in the TEFSM meter and can create a 2-D pseudo-section of EPD along the traverse line [23, 25]. Anomalies can be related to underlying geological and or hydrogeological properties to identify the groundwater potential. The study [23] also provides an option to send data to analysis by their experts.

The other approach involves downloading the data, plotting the EPD against depth, and correlating it to drilling data or the general hydrogeology of the area to delineate potential aquifers. Gomo proposed this idea [15], and it has since been used in several studies.

4. The emergence of TEFSM geophysical technology

For many groundwater professionals, the emergence of TEFSM technology in the industry has been confronted with several questions and sometimes doubts. This is mainly due to the limited information made available about the working principles and guidance on application. Due to this situation, the TEFMS equipment is already widely in use, accompanied by limited technical guidance. However unconventional as it sounds, it is what it is. It is therefore inevitable to ask a few questions. How did we get to this situation and could this be the new reality of the modern world? No one may know the exact factors that contributed, but it is an important discussion to have among groundwater professionals. While there could be other factors that might have led to this situation, the real need for cost-effective alternative groundwater exploration geophysical techniques has a huge influence.

There is many conventional geophysical groundwater prospecting equipment, but high cost is a prohibitive factor in many developing regions [12, 13, 15, 26, 27]. At the same time, one can also argue that expensive equipment does not always translate to successful groundwater exploration. Hence, the need for cost-effective alternatives

will always remain. Many practitioners/contractors acquired and used the TEFSM equipment with little to no understanding of the operating principles of what it measures, and even its units or limitations. For example, several researchers suggest that the TEFSM approach measures resistivity [28–30].

Just as the saying goes, “Necessity is the mother of all inventions,” due to the need and the affordability of the equipment, many practitioners and drilling contractors were, therefore, willing to give it a try. With groundwater prospecting equipment, it is possible to assess if a tool is good for its purpose or not, because drilling results will show. For further insights, survey data can be corroborated by drilling data. Revelations from the published studies show that the TEFSM can delineate aquifers, for example, in Refs. [11, 12, 14, 15, 27, 31]. The sudden uptake of the technology is, therefore, a reflection of its capabilities, irrespective of the limited understanding of the principles behind the technique.

5. Challenges

Although the TEFSM has been successfully applied in several hydrogeology environments, some challenges need to be highlighted and discussed. The main challenges relate to the lack of technical guidance for analysing and interpreting the TEFSM data and a limited understanding of the noise effects on TEFSM measurements.

5.1 Access to information about the TEFSM approach

There is very little information accessible in the public domain about the TEFSM techniques. The manuals, for example [23], typically lack the scientific background behind the approach and measurement parameters, and in some instances, measurement units are not specified, as recently noted [27]. Some of the equipment, such as ADMT-300S Underground water detector, does not display the measurement units on the screen [23] or in the manual. It is due to this lack of information about the products that raises questions.

One could argue that it has been a nightmare. You decide to test a piece of equipment, and it gives good results, but at the same time, one is not entirely sure how it works or perhaps what it is measuring. The manuals do not elaborate on quality control issues, analysis, and interpretation. There is fragmented information about the processing of the data acquired through inbuilt software that creates 2D and or 3D models either on the equipment or a cell phone. There are some illustrations showing groundwater drilling targets on the 2D models, but they are not accompanied by the meaningful hydrogeological explanations. The alternative approach of sending data for analysis by their experts has some major limitations. Groundwater practitioners should be able to self-analyse their data and relate it to site hydrogeology settings to optimise the understanding. This is important for professional development because it builds experience and competence in the method of investigation.

Due to the lack of information about this approach, various published studies have used the TEFSM in different earth science fields with limited knowledge of the principles behind the equipment [17, 28, 30, 32]. This is evidenced by the absence of any theoretical background of the principles for the equipment used. In most of these studies, only the equipment model name was mentioned, without giving details of the underlying principles, for example on the PQWT – TC 150 [17, 33, 34] and the

ADMT-300S [28, 35]. Despite the lack of scientific information about the TEFSM equipment, all these studies reported various successes in groundwater exploration.

In 2023, several publications relating to TEFSM have emerged. Papers relating to the development of theoretical understanding of the TEFSM anomalies mainly come from the Republic of China, where the approach originated [10, 31]. Considering the references used in these papers, it is evident that the TEFSM approach has evolved over several decades [9, 36]. However, most of the sources relating to the development of the Frequency selection method approach are generally not accessible in the public domain, and some of them are only available in one language, for example, Yang et al. [31]. It is anticipated that with improved access to published research on the approach, both theoretical and practical understanding will improve to optimise the application of the technology.

5.2 Analysis and interpretation of TEFSM data

Although the TEFSM is already being applied in many parts of the world, there is currently no standardised approach to analyse and interpret the TEFSM data for groundwater exploration purposes, the challenge that can hinder the meaningful application of the approach. Due to this knowledge gap, researchers have been using different approaches to analyse and interpret TEFSM data in hydrogeology applications.

Studies such as [12, 13, 17, 18, 37] have used the software provided by the equipment developer to analyse the EPD data from the TEFSM measurements. The software can be installed on a smartphone or computer and typically produces 2D and 3D subsurface models. From these models, the low EPD anomalies are identified vertically and laterally to locate potential drilling targets. For example, in a study to explore the “geothermal water of the Maoyanhe hot Spring in Zhangjiajie,” [12] indicated: “As can be seen from Figure 6b, the potential difference curve near 45 m of measurement line 7 appears as the “hanging noodle” phenomenon. The contours in depths 0 ~ 125 m generally bend downward, the contours in depths 125 ~ 200 m generally bend upward, and the contours in depths above 200 m tend to bend downward. It is inferred that there is groundwater in the shallow part of the anomaly within 120 m, and the depth of the deep water may be 200 m later.” **Figure 2** shows an analysis of the TEFSM data using bipolar profile and pseudo-section diagrams [12].

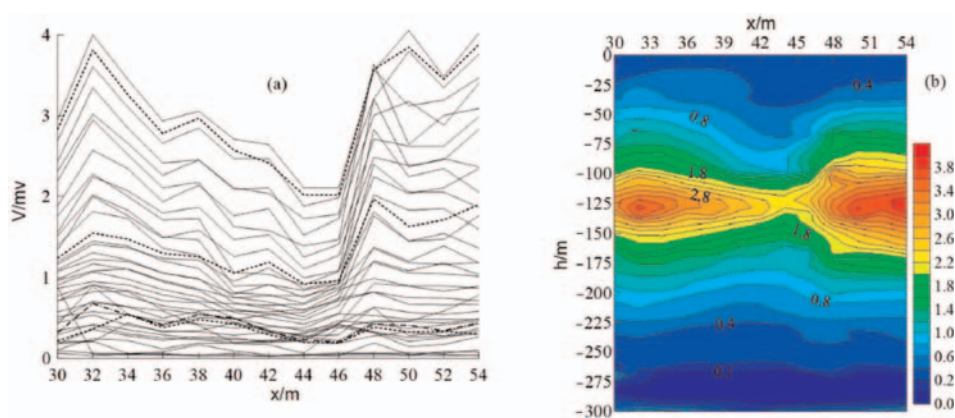


Figure 2.
 Analysis of TEFSM data using bipolar profile and pseudo-section diagrams [12].

Although studies are using the software provided by the equipment developers to analyse the TEFSM data, there are several challenges associated with this approach. Firstly, no EPD values or range have been defined to constitute a good groundwater potential target for drilling purposes. Without some criteria to classify the groundwater potential based on EPD values, it is not meaningful to refer to low EPD areas of the profiles as the potential targets for drilling. The vital question of how low the so-called “low EPD contrast” remains an important one. This information is required to guide practitioners in groundwater exploration work. This is despite that both the electric and magnetic fields will likely vary in space and time, therefore influencing the measurements. Secondly, these studies do not provide any principles about the software used to model the data and to create the pseudo sections in their reporting. A good understanding of the principles, strengths, and limitations of software is important to optimise its application and future references. This observation suggests that the software accompanying the TEFSM equipment is being used without consideration of the science behind it. This, too, is problematic because creating a pseudo-section involves modelling, which may introduce an unknown source of uncertainty in contrast to directly analysing the raw EPD profiling data.

Some studies, such as [14, 15, 28, 38], have focused on both horizontal and vertical profiles of EPD, with the emphasis being on the latter. The vertical profile data from the measurement stations is plotted on the graph, with depth on the y-axis and EPD on the x-axis. Depths with low electric potential difference contrast anomalies are identified as the potential aquifers and targeted for drilling (**Figure 3**). Low EPD values corresponding to the local aquifer are defined based on knowledge gained from previous experience working in the area. The approach has potential, but the main challenge is to ascertain the EPD values corresponding to the aquifer. This is the major source of uncertainty that needs to be addressed. The challenge can be reduced by using hydrogeology data (e.g., geological logs, aquifer depths/water strikes, and yields) from existing wells or test wells to calibrate the EPD TEFSM data before use at local scales. For specific geology, calibration of the EPD geophysical data will define the EPD range associated with aquifers at a local scale. Despite the reported success, this approach needs improvement.

5.3 Noise effects on TEFSM measurements

Lack of knowledge about the noise effects on the TEFSM measurements is also an obstacle. The manuals accompanying the equipment do not provide any information about the types of noise effects, and how they can be identified and filtered or corrected before data analysis. Any geophysical equipment is inevitably vulnerable to noise effects, and the TEFSM cannot be an exception. **Figure 4** shows the TEFSM profile with noise effect between stations 10–16 m along the survey line length, which was encountered at the research site for the Institute for Groundwater Studies at the University of Free State. The data was collected using ADMT-300S Groundwater Detector. The low EPD anomaly from stations 10–16 m was thought to be because of buried electrical cables for road lights. However, without guidance on what different noise effects can affect the TEFSM measurements and the nature of their effects, the same anomaly can be easily mistaken for geological discontinuities and therefore groundwater potential. Knowledge of the effect of noise on TEFSM must be investigated and conceptualised.

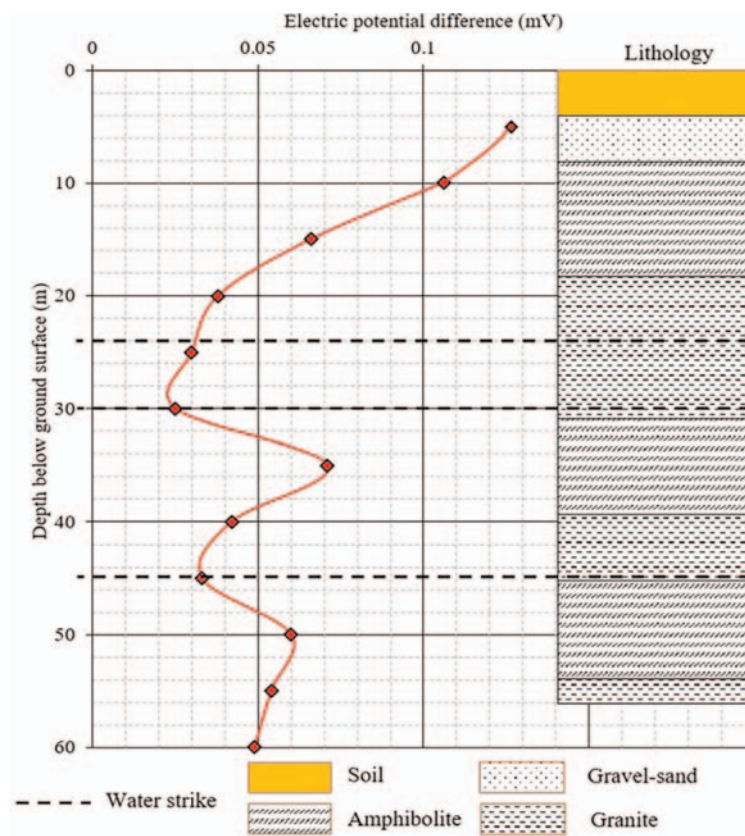


Figure 3.
 Example of granite aquifer lithology delineated based on the low EPD contrast depth anomalies [38].

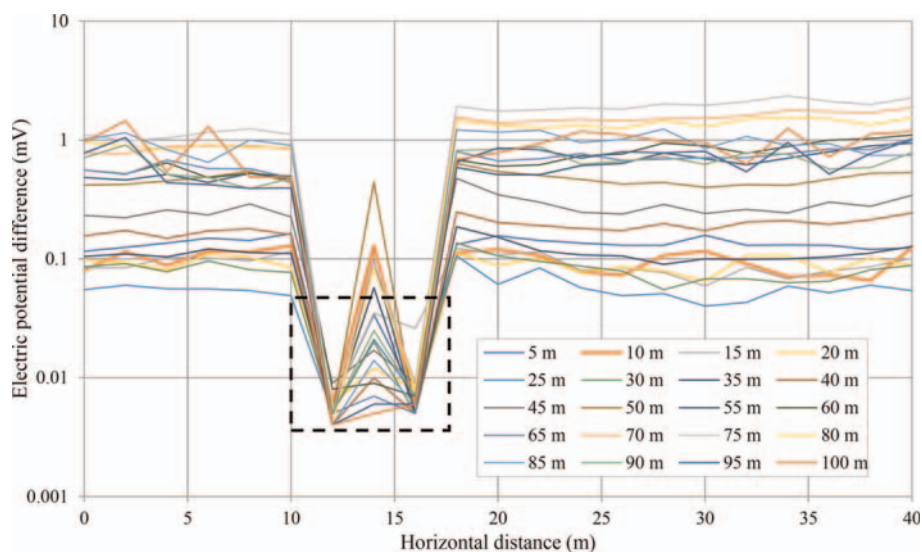


Figure 4.
 Horizontal EPD profile with noise effect between 10 and 16 m, and data was collected using ADMT-300S Groundwater Detector.

6. Positives

When compared to conventional surface geophysics techniques such as electrical resistivity, the TEFSM has several appealing capabilities. The cost of TEFSM equipment is much lower; it is for this reason that TEFSM has been popularly received in developing regions such as Africa [13–15, 27]. With the TEFSM approach, it is faster to acquire measurements, which reduces the cost of field investigations. The TEFSM equipment is portable to handle and operate; for example, basic TEFSM equipment, such as TC 150 or ADMT equipment, only requires a minimum of 10 m electrode spacing to enable a vertical profile measurement. The real timeliness of the TEFSM enables in-field decision-making to assess the groundwater potential of sites and to prioritise sites for detailed investigation [14].

The other important feature of the TEFSM that is noteworthy is the depth capabilities. TEFSM equipment with a depth investigation capacity of 4000 m is now available on the market, for example, (PQWT-L4000 and ADMT-4000SX-32D 32). However, reported studies have thus far shown the success of the TEFSM approach in delineating aquifers at depths up to 500 m, and this depth covers borehole depths for typical day-to-day groundwater exploration activities. Studies such as [26] successfully delineated a sandstone aquifer between 300 and 350 m deep in Saudi Arabia using the ADMT-9 New Generation Portable Detector. Using the ADMT-300S Groundwater Detector, Gomo [15] applied the TEFSM approach to delineate layered sandstone aquifers up to 150 m depth in the Karoo region of South Africa.

A geothermal water exploration of hot springs in marine carbonate rocks in Zhangjiajie was conducted, where borehole logs and drilling data up to 500 m depth were corroborated with EPD data from the TEFSM [12]. In another study, Yulong et al. [10] investigated the application of TEFSM on 131 boreholes between depths 75.2–42.8 m during a rural drinking water supply project in Guangxi Province, China, where they concluded that the TEFSM geophysical approach was suitable for shallow groundwater exploration. Recently, Gomo and Ngobe [38] successfully applied the TEFSM to groundwater supply boreholes up to 100-m deep in a typical granite aquifer in Eswatini, Southern Africa [14], and further delineated a dolomite aquifer from 170 to 210 m in South Africa and Southern Africa.

7. Discussion

It is also important to highlight that the fact that scientific understanding is probably behind the pace of uptake and field application of the TEFSM geophysical equipment does not make the technology fake, unworthy, or fraudulent, as some have claimed. Given below is an example from an online forum where the legitimacy of the TEFSM groundwater detector equipment was being questioned. For example, from Ref. [39], *“There are several legitimate geophysical methods to look for groundwater and a number of fraudulent instruments sold as water detectors. There is no such instrument as a water detector, so if someone tries to sell you something called a water detector you should run away quickly.”*

What is needed is to promote health discussions and research to improve the understanding of theoretical and field applications of the TEFSM techniques. Through health discussions, the sharing of case studies where these techniques have been

applied can be stimulated. Branding the TEFSM tools as “unproven or fake” discourages the sharing of data and information from TEFSM applications as practitioners might fall victim to being labelled unprofessional or fake as well. The impacts also extend to peer-reviewed scientific journals, where work from the TEFSM approach can be easily dismissed based on legitimacy without giving due attention to the actual science. There is also another major risk of creating a polarised academic and research world, which is not needed. Despite the challenges, studies are showing the capabilities of the TEFSM approach to delineate aquifers and other groundwater investigations such as contamination and geothermal investigations. The best that researchers and professionals can do is to promote frank discussions and judgements based on experience and grounded facts and encourage more research to improve the understanding and provide groundwater solutions to global problems.

8. Conclusions

The study discusses emerging issues on the application of the recently developed TEFSM geophysical approach in groundwater exploration. The TEFSM approach has been favourably received in many developing regions but is also starting to receive attention in some of the developed regions. However, the pace of uptake of the technology appears to have outpaced the scientific understanding required for the meaningful application of the approach. The sudden uptake of the TEFSM can be attributed to the need for low-cost and effective groundwater exploration geophysical approaches. Quick real-time measurements, portability, and deeper exploration ability are other key attributes of the TEFSM approach. The main challenges relate to a lack of information about the principles of the approach and the absence of standard guidelines for data analysis and interpretation. The literature presents two concepts on the principles of the TEFSM approaches. Some studies suggest that the TEFSM measurements are mainly controlled by subsurface resistivity, while others attribute it to the shielding effectiveness. Research to refine the theory and field applications to the performance of the TEFSM approach is required. The development of standard approaches to analyse and interpret TEFSM data and understanding the noise effect on field measurements is a priority to optimise the application of the TEFSM. Health professional debate is encouraged to promote more research and the open sharing of knowledge to optimise the application of the approach in groundwater investigations.

Conflict of interest

The author declares that they have no conflict of interest.

Author details

Modreck Gomo* and Thandeka Ngobe
Institute for Groundwater Studies, Faculty of Natural and Agricultural Sciences,
University of the Free State, Bloemfontein, South Africa

*Address all correspondence to: modgomo@yahoo.co.uk

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